

Square Point Infra IIT KGP

4 Questions:

- 1 DevOps (Linux)
- 1 Python
- 2 REST API

DevOps

We were given many log files (name was access.<number>.log for nginx server).

We have to filter the log entries with “FAILURE” as the status and sort them based on the time stamp, latest being at the top. If two entries timestamp matches, then sort them based on the response time.

Then select the top 50 from the final output and store in a file, location was mentioned in the question.

Python

2. Function Composition

Create a function named *compose* that:

- Accepts any number of functions as arguments
- Returns a new function representing their composition
- Passes arguments to the first function
- Passes each function's output to the next function
- Returns the final output

The provided functions are:

- *add(*args)*: Returns the sum of all arguments
- *square(a)*: Returns the square of a number
- *splitter(a)*: Divides a number by 2 and returns $\lfloor \frac{a}{2} \rfloor$
- *my_max(a)*: Returns the maximum value in a number or list
- *my_min(a)*: Returns the minimum value in a number or list

Note: Only the first function handles multiple arguments; subsequent functions take a single argument.

Example
`functionsList = [add, splitter]`

The function call `compose(functionList[1], functionList[2])` should return a function, call it `composedFunctions`. If `argumentsList = [2, 3]`, `composedFunctions(2, 3)`, should return `[2, 3]`.

- `add(2, 3) -> 5`
- `splitter(5) -> [2, 3]`

```
1 > #!/bin/python3...
37 #
38 # Complete the 'compose' function below.
39 #
40 # The function is expected to return a FUNCTION.
41 # The function accepts following parameters:
42 # 1. ARGUMENTS *functionsList
43 #
44
45 def compose(*functionsList: Callable) -> Callable:
46     ... # Write your code here
47     for func in functionsList:
48         if func == my_max:
49             return my_max()
50     ...
51
52
53 > if __name__ == '__main__': ...
```

Line: 37 Col: 2

Test Results

Custom Input

Run Code

Run Tests

Submit

REST API

Questions similar to this.

<https://github.com/sumantopal07/Rest-API-Intermediate-Hackerrank-Test>

In one, we had to find the total votes.

In another, some “rating” need to be calculated based on the formula mentioned in the question.